

return forecasts against risk. The key to generating alpha is forecasting. That requires superior information or superior ability to process public information.

While the fundamental law was intended originally as a portfolio construction tool, Knut Kjær, the founding CEO of Norges Bank Investment Management, which manages the Norwegian sovereign wealth fund, also used the intuition of the fundamental law as a management style. Since alpha is created by finding many different forecasts that are independent, Kjaer's philosophy was to delegate widely to specialized managers. He tried to ensure that they act as independently as possible. He explained:⁴

To achieve the greatest possible level of independence for decisions on active investments, these must be delegated to many different groups and individuals. This delegation must be real, without intervention from superiors, provided that the individual employee stays within the agreed structure and risk limits. . . . The sense of ownership which derives from responsibility and an absence of intervention from superiors is an important motivation and driving force for the development of this expertise.

There are two very important limitations of the fundamental law. The first is that *ICs* are assumed to be constant across *BR*. The first manager whom you find may truly have a high *IC*, but the one hundredth manager whom you hire probably does not. As assets under management increase, the ability to generate *ICs* diminishes. Indeed, the empirical evidence on active management, which I cover in Part III of this book, shows *decreasing returns to scale*: as funds get bigger, performance deteriorates. This effect is seen in mutual funds, hedge funds, and private equity.⁵ Thus, *ICs* tend to fall as assets under management rise.

Second, it is difficult to have truly independent forecasts in *BR*. Manager decisions tend to be correlated and correlated bets reduce *BR*. An equity manager with overweight positions on 1,000 value stocks offset by underweight positions in 1,000 growth stocks has not placed 1,000 different bets; he's placed just one bet on a value-growth factor. Hiring one hundred different fixed income managers who are all "reaching for yield" by buying illiquid bonds gets you not one hundred different bets but rather a single bet on an illiquidity factor. Correlated factor bets tend to dominate at the overall portfolio level—a reason why top-down factor investing is so important (see chapter 14).

Despite its influence in industry, Grinold's fundamental law makes only scant appearance in the academic literature. This is because Grinold's framework is a statistical model devoid of economic content. It says nothing about where positive

⁴ NBIM Annual Report, 2007, "Ten Years of NBIM."

⁵ See Chen et al. (2004), Fung et al. (2008), and Kaplan and Schoar (2005), respectively.